

Experiment 9: Introduction to Azure DevOps: Overview of Azure DevOps Services, Setting Up an Azure DevOps Account and Project

Overview of Azure DevOps

Azure DevOps is a comprehensive suite of cloud-based services designed to support the entire software development lifecycle. It provides tools for planning, developing, testing, delivering, and monitoring applications. Here are the primary services offered:

- **Azure Repos:**

A set of version control tools that allow you to host Git repositories or use Team Foundation Version Control (TFVC). It offers collaboration features such as pull requests, branch policies, and code reviews.

- **Azure Pipelines:**

A CI/CD service that helps automate builds, tests, and deployments. It supports multiple languages, platforms, and can run on Linux, Windows, or macOS agents.

- **Azure Boards:**

A work tracking system that helps teams manage work items, sprints, backlogs, and Kanban boards. It facilitates agile planning and reporting.

- **Azure Test Plans:** Provides a solution for managing and executing tests, capturing data about defects, and tracking quality.

- **Azure Artifacts:**

Allows you to create, host, and share packages (such as Maven, npm, NuGet, and Python packages) with your team, integrating package management into your CI/CD pipelines.

These services integrate with each other and with popular third-party tools to create a cohesive DevOps ecosystem.

Setting Up an Azure DevOps Account

Before you can start using Azure DevOps services, you need to set up an account and create an organization. Follow these steps:

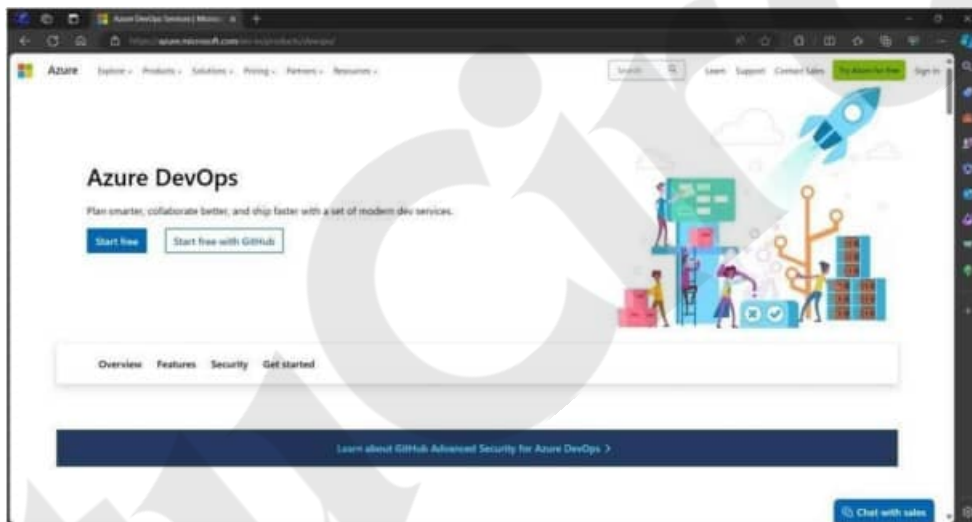
Step 1: Sign Up for an Azure DevOps Account

1. Open Your Web Browser:

- Navigate to the Azure DevOps website: <https://dev.azure.com>.

2. Sign In or Create a Microsoft Account:

- If you already have a Microsoft account (such as Outlook, Hotmail, or Office 365), click “Sign in”.
- If you do not have a Microsoft account, click “Create one!” and follow the instructions to create a new Microsoft account.



3. Accept the Terms and Conditions:

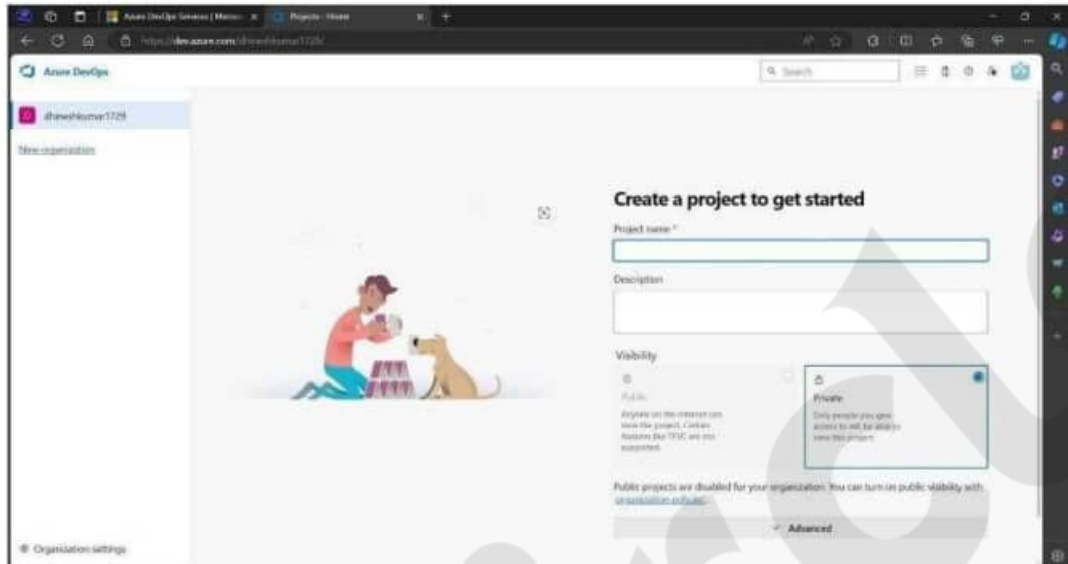
- Review and accept the terms if prompted.

Step 2: Create an Azure DevOps Organization

1. Create a New Organization:

- Once signed in, you will be prompted to create an Azure DevOps organization.
- Enter a unique name for your organization (e.g., YourCompanyDevOps or MyPersonalOrg).

- **Select a Region:** Choose the geographic region where your data will be stored (select the one closest to you for optimal performance).
- Click “**Continue**” or “**Create**”.



2. Review Your Organization's Dashboard:

- Once created, you will see an overview dashboard for your organization. This dashboard provides navigation links to Repos, Pipelines, Boards, Test Plans, and Artifacts.

3. Creating an Azure DevOps Project

After setting up your organization, the next step is to create a project. A project in Azure DevOps is a container for all your source code, pipelines, boards, and other resources.

Step 1: Create a New Project

1. Navigate to “New Project”:

- On your organization's dashboard, click the “**New Project**” button.

2. Configure Your Project:

- **Project Name:** Enter a descriptive name for your project (e.g., HelloDevOps).
- **Description:** Optionally, provide a brief description (e.g., “A sample project to demonstrate Azure DevOps services”).
- **Visibility:**

- Choose “**Private**” if you want to restrict access to your project.
- Choose “**Public**” if you are okay with the project being accessible to anyone.
- **Advanced Options (Optional):** You can choose a version control system (Git is the default) and a work item process (Agile, Scrum, or Basic). For most beginners, the defaults are recommended.
- Click “**Create**”.

Step 2: Explore Your Project Dashboard

1. Project Overview:

- Once your project is created, you will be directed to the project dashboard.

Here you will see navigation options for:

- **Repos:** Where your code is stored.
- **Pipelines:** For build and release automation.
- **Boards:** For work tracking and agile planning.
- **Test Plans:** For managing and running tests.
- **Artifacts:** For hosting packages.

2. Familiarize Yourself with the Interface:

- Click through each section (e.g., Repos, Pipelines, Boards) to get a sense of the available features.

